Mohamed Ahmed

Fullstack Engineer

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SUMMARY

Fullstack Engineer with hands-on experience building scalable web applications and backend systems, along with modern, responsive user interfaces. Skilled in designing robust backend architectures, crafting APIs, and managing complex frontend/backend integrations. Built a production microservices platform at CareComm, including its APIs and third-party integrations. Adept at writing clean, maintainable code and delivering reliable, high-performance solutions. Recently contributed to full-stack projects focused on scalability, performance, and seamless user experience.

SKILLS

Frontend: React, Next.js, TypeScript, Tailwind CSS, Bootstrap, jQuery, JavaScript, HTML, CSS, Redux Toolkit, Apollo Client, TanStack Query

Backend: Node.js, Express.js, GraphQL (Apollo Server), REST APIs, WebSockets, Stripe, MCP (Model Context Protocol)

Databases: MongoDB

AI & Data: Python, NumPy, pandas, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), vector and hybrid search (BM25), LLM APIs (Gemini, OpenRouter)

EXPERIENCE

Software Engineer, Full-Time | *CareComm* *Jul 2025 – Present*

- Built and scaled a microservices architecture using Node.js, powering a dashboard that automates website creation and social media management for marketing clients.
- Developed the backend server as a modular, domain-based platform with independent modules for Social, Ads, Messaging, Appointments, Support and an agent gateway, sharing a common core for auth, permissions and error handling.
- Designed and maintained a GraphQL API (Apollo Server) with DataLoader batching, query depth and complexity limits, and real-time subscriptions over WebSockets (graphql-ws) to unify data across internal services and third-party platforms.
- Automated end-to-end social media workflows (post scheduling, publishing and analytics aggregation) across Facebook, Instagram, X and TikTok, boosting marketing team productivity and data accuracy.
- Integrated AWS services (S3 for file storage, SES for email, SNS for notifications), Stripe for billing, and WhatsApp Business messaging.
- Built the Social Management frontend, a React 19, Vite and TypeScript single-page app with Apollo Client, TanStack Query and Redux Toolkit, Tailwind CSS v4, and full English/Arabic (i18n) support.
- Delivered rich management screens: campaigns, ads and audiences, content drafts and publishing calendar (FullCalendar), comments inbox, analytics dashboards (Recharts), lead forms and lead management, and billing with Stripe.
- Built drag-and-drop workflows (dnd-kit), data tables (TanStack Table), and file previews/exports (XLSX, PDF, DOCX) for marketing teams.

Full Stack Developer, Freelance | *Remote* *Dec 2024 – Present*

- Built and maintained full-stack web applications using Next.js, React and Node.js.
- Developed RESTful APIs and backend services with Django and Express, integrated with MongoDB.
- Managed deployments using Docker, version control with Git.

EDUCATION

Bachelor's degree in Science, Al-Azhar University — Sep 2017 – Jul 2022

PROJECTS

Republic Burger – Web Application (Food Ordering System) GitHub Backend | Live Demo | GitHub Frontend

- Built a full-featured web application with user authentication, project showcase, and rating functionality using Node.js.
- Designed and integrated a MongoDB database, implemented RESTful API endpoints, integrated Stripe for payment processing and served static files using Express for production-ready deployment.
- Developed a responsive frontend using Next.js with Tailwind CSS for UI.

Buy Ecommerce – Full-Featured E-commerce Platform GitHub Link | Live Demo

- Built a responsive e-commerce frontend using React, Vite, and Tailwind CSS.
- Integrated with backend APIs for dynamic product browsing and data handling.
- Implemented routing, forms, image galleries, carousels, and notifications for a smooth user experience.

Portfolio – Personal Portfolio Showcasing Projects and Skills GitHub Link | Live Demo

- Built a personal portfolio showcasing my skills and projects.
- Developed a responsive frontend using Next.js, GSAP animation, TypeScript and Tailwind CSS.

Lumio – AI-Powered News Chat Platform GitHub Frontend | GitHub Backend | GitHub AI Service

- Built a RAG service in Python that scrapes BBC and RSS sources, cleans and chunks articles, generates embeddings, and answers grounded questions with source citations.
- Implemented hybrid retrieval combining BM25 keyword search with MongoDB vector search, plus query preprocessing and a streaming chat endpoint.
- Backend (Node.js, Express, GraphQL): JWT authentication, Stripe payments, transactional email, and a streaming chat route connected to the RAG service.
- Frontend (React, Vite, TypeScript): chat interface with streamed Markdown answers, source citation cards, TanStack Query and React Hook Form.
- Stack: Python, MongoDB, Gemini and OpenRouter LLMs, scheduled ingestion (APScheduler), Render deployment.

CERTIFICATIONS

Frontend Professional Diploma – Route Certification

- HTML, CSS, JavaScript, Bootstrap, jQuery, Tailwind CSS, TypeScript, React and Next.js.

Backend Diploma – Jonas Schmedtmann (Udemy)

- Node.js RESTful APIs and web apps with authentication, security and payments.

AI & Data Science Track – Route GitHub Assignments

- Python, NumPy, pandas, Natural Language Processing (NLP) and Retrieval-Augmented Generation (RAG).
- Completed 10 hands-on assignments covering EDA, KNN, logistic regression, SVM, ensemble methods, artificial neural networks (ANN) and convolutional neural networks (CNN) on real datasets using NumPy, pandas and scikit-learn.